

PIETER AYAL

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EDUCATION

University of Manchester

Manchester, United Kingdom

Bachelor of Science (Honours) in Computer Science with Industrial Experience

Sep. 2024 – Jun. 2028

- Academic Performance (First Class): First Year: **75.3%** | Second Year: **76.8%**
- Relevant Coursework: Data Structures and Algorithms, Compilers, System Architecture, Concurrent Programming, Distributed Systems, Operating Systems, Software Engineering

EXPERIENCE

Amazon Web Services

Dublin, Ireland

Software Development Engineering Intern (Incoming) | Java

Jan. 2027 – Sep. 2027

- Joining the **Control Plane** team for **AWS ElastiCache**, Amazon's managed in-memory caching service.

SLB

Abingdon, United Kingdom

HPC Software Engineering Intern (Incoming) | C++

Sep. 2026 – Dec. 2026

- Upscaling **embedded meshes** for **HPC simulation** workflows.

Thought Machine

London, United Kingdom

Software Engineering Intern | TypeScript, React, REST

Jun. 2026 – Sep. 2026

- Architected and shipped a full-stack **Balances Visualiser**, replacing manual log and query inspection as the first step in diagnosing balance discrepancies for on-call engineers, and adopted as onboarding material for the ledger's accounting model.
- Drove the design end-to-end, authoring an RFC reviewed by **20 engineers across 4 teams** and landing the implementation across **30 merged PRs** through cross-team code and technical review.
- Built a **parsing layer** validating financial transaction payloads against the internal banking engine's REST API schema, rejecting malformed and non-conformant data before it reached the ledger.

PROJECTS

EMEA Internship Alerting Bot | Go, SQLite, Docker, GitHub Actions | github.com/pieter124/senior-intern-bot

- Built a zero-human-in-the-loop internship alerting discord bot polling multiple **ATS APIs** (Greenhouse, Ashby) behind one Poller interface, with per-source DTOs and mappers so incompatible wire formats never reach the domain model.
- Designed a **three-way classifier** (accept / review / reject) in which *unknown* is the zero value and absence of evidence can never reject, making silent drops unreachable by construction rather than by test coverage.
- Eliminated an **81 MB** per-cycle fetch (**5.7 GB/day** across 33 boards) after establishing classification reads only `title` and `location`; ordered persistence after delivery so an undelivered posting is never marked seen.

Sharded Concurrent Ledger | Go, gRPC, Protobuf | github.com/pieter124/concurrent-bank-ledger

- Designed and benchmarked three concurrency strategies: fine-grained per-account mutexes (**~1450 ns/op**), a single-actor model (**~2225 ns/op**), and a lock-free sharded-actor architecture (**~1080 ns/op**).
- Implemented a two-phase **reserve/credit/finalise** protocol for cross-shard transfers, ensuring zero-sum conservation under concurrent load, with a **write-ahead log** providing crash recovery and idempotent state rebuilding on restart.

EXTRA-CURRICULAR ACTIVITIES

Technical Interview Coach | Manchester AI Club, University of Manchester

- Ran **50+** DSA mock interviews for students, several of whom went on to offers at Google, Amazon and JPMorgan.

Google Student AI Hackathon 2026 | 3rd Place | github.com/pieter124/google-ai-hackathon-2026

Hackabot (Robotic Arm Track) 2026 | 2nd Place | github.com/pieter124/hackabot26

SAT Solver Competition 2025 | 2nd Place | github.com/pieter124/rustSAT

HPC Summer School 2025 | Edinburgh Parallel Computing Centre, University of Edinburgh

TECHNICAL SKILLS

Languages: Go, C++, C, TypeScript, Rust, Java, Python

Frameworks & APIs: React, gRPC, Protobuf, REST

Infrastructure: Docker, SQLite, GitHub Actions, Linux, Git

HPC: OpenMP, MPI, SLURM